

Lab 7– Convolution-II *

EE252L: Signals and Systems Lab

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Task 1

```
%Read image, visualize it, and display its size and class.  
I = imread("flower.jpg");  
% Convert to double precision  
I = im2double(I);  
% Display  
figure,  
imshow(I)
```



```
whos I
```

Name	Size	Bytes	Class	Attributes
I	500x351x3	4212000	double	

```
% Split the image into the 3 component color intensity planes or matrices.
```

```
[R, G, B] = imsplit(I);
```

```
whos R
```

Name	Size	Bytes	Class	Attributes
R	500x351	1404000	double	

```
% Create a 2-D convolution filter or kernel for averaging or smoothing
```

```
W = 5;
```

```
h = ones(W)./W.^2;
```

```
%Convolve R with h to blur the image
Ravg = conv2(R,h,"same");
% Convolve G with h to blur the image
Gavg = conv2(G,h,"same");
% Convolve B with h to blur the image
Bavg = conv2(B,h,"same");
% Concatenate the three matrices to put together the color image
Iavg = cat(3,Ravg,Gavg,Bavg);
whos Iavg
```

Name	Size	Bytes	Class	Attributes
Iavg	500x351x3	4212000	double	

```
imshow(Iavg,[])
```



```
%Read image, visualize it, and display its size and class.  
I = imread("flower.jpg");  
% Convert to double precision  
I = im2double(I);  
% Display  
figure,  
imshow(I)
```



```
whos I
```

Name	Size	Bytes	Class	Attributes
I	500x351x3	4212000	double	

```
% Split the image into the 3 component color intensity planes or matrices.  
[R, G, B] = imsplit(I);  
whos R
```

Name	Size	Bytes	Class	Attributes
R	500x351	1404000	double	

```
% Create a 2-D convolution filter or kernel for averaging or smoothing
W = 10;
h = ones(W)./W.^2;

%Convolve R with h to blur the image
Ravg = conv2(R,h,"same");
% Convolve G with h to blur the image
Gavg = conv2(G,h,"same");
% Convolve B with h to blur the image
Bavg = conv2(B,h,"same");
% Concatenate the three matrices to put together the color image
Iavg = cat(3,Ravg,Gavg,Bavg);
whos Iavg
```

Name	Size	Bytes	Class	Attributes
Iavg	500x351x3	4212000	double	

```
imshow(Iavg,[])
```



1) Increase the filter size W . What is the effect of increasing the size of the filter?

Each output pixel is the average of a $W \times W$ neighborhood. A larger W means each pixel averages over more neighbors, so the image gets blurrier

2) Look closely at the boundaries of the output image. What happens to the border pixels? Why?

With `conv2`, the output is the same size as the input, but near the edges the kernel window hangs off the image. MATLAB assumes the pixels outside the image are zero. So at a border pixel, part of the window includes imaginary black (0) pixels, and the average gets pulled toward zero.

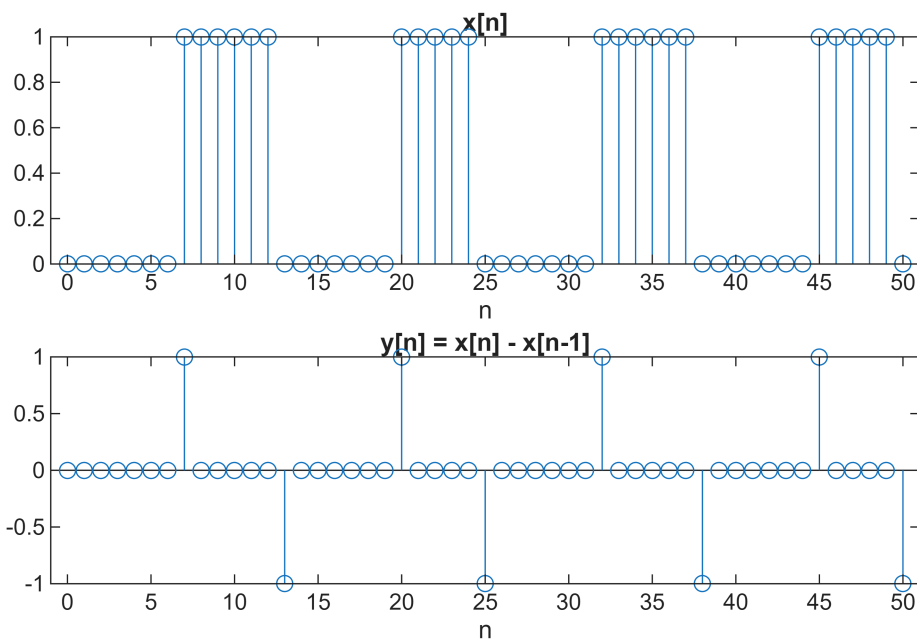
Task 2

```
n = 0:50;
```

```
x = double((sin(0.5*n) + 0.2) < 0);
```

```
h = [1 -1];  
y = conv(x, h);  
y = y(1:length(x));
```

```
figure  
subplot(2,1,1)  
stem(n, x)  
title('x[n]')  
xlabel('n')  
subplot(2,1,2)  
stem(n, y)  
title('y[n] = x[n] - x[n-1]')  
xlabel('n')
```



Task 3

```
load echart.mat;  
figure;  
imshow(echart)
```


 E W S X
 E W S X M
 E W S X M P

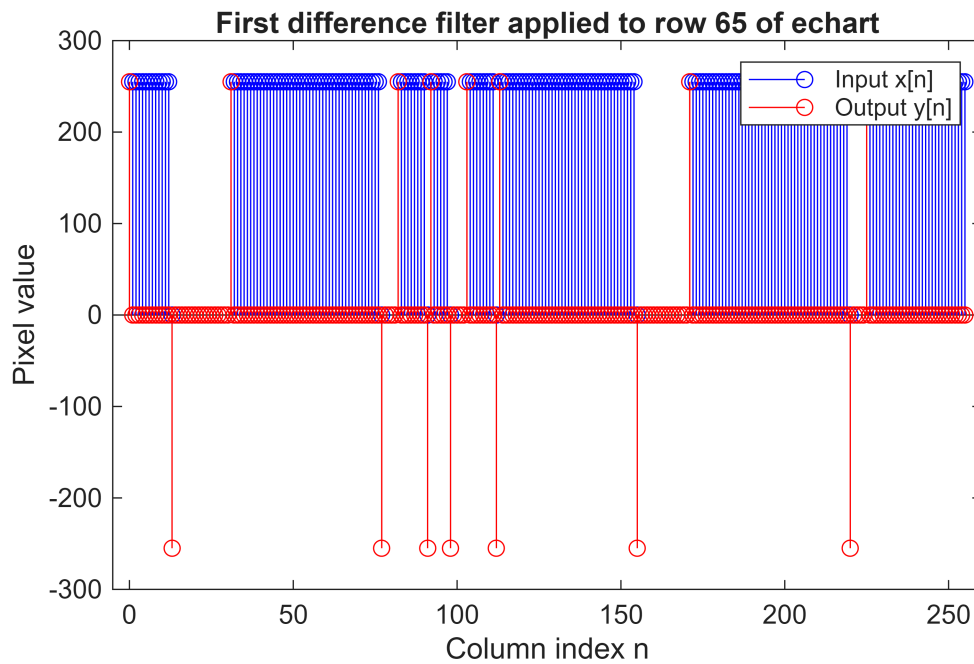
```
figure;
imshow(echart(1:65,:))
```


 E W C V

```
row = double(echart(65,:));
n = 0:length(row)-1;

h = [1 -1];
y = conv(row, h);
y = y(1:length(row));

figure
stem(n, row, 'b')
hold on
stem(n, y, 'r')
hold off
title('First difference filter applied to row 65 of echart')
xlabel('Column index n')
ylabel('Pixel value')
legend('Input x[n]', 'Output y[n]')
```



In the input signal, 0 represents a black pixel (part of a letter stroke) and 255 represents a white pixel (the background), so the row is mostly white with short black runs wherever the scan line crosses a letter. In the output signal $y[n] = x[n] - x[n-1]$, a value of 0 means neighboring pixels are the same, so the scan is inside a flat region that is all white or all black. A value of -255 means the pixel went from white to black, which marks the left edge of a letter stroke, and $+255$ means it went from black to white, which marks the right edge. So the non-zero outputs mark the edges, and the sign shows the direction of the change.

Task 4

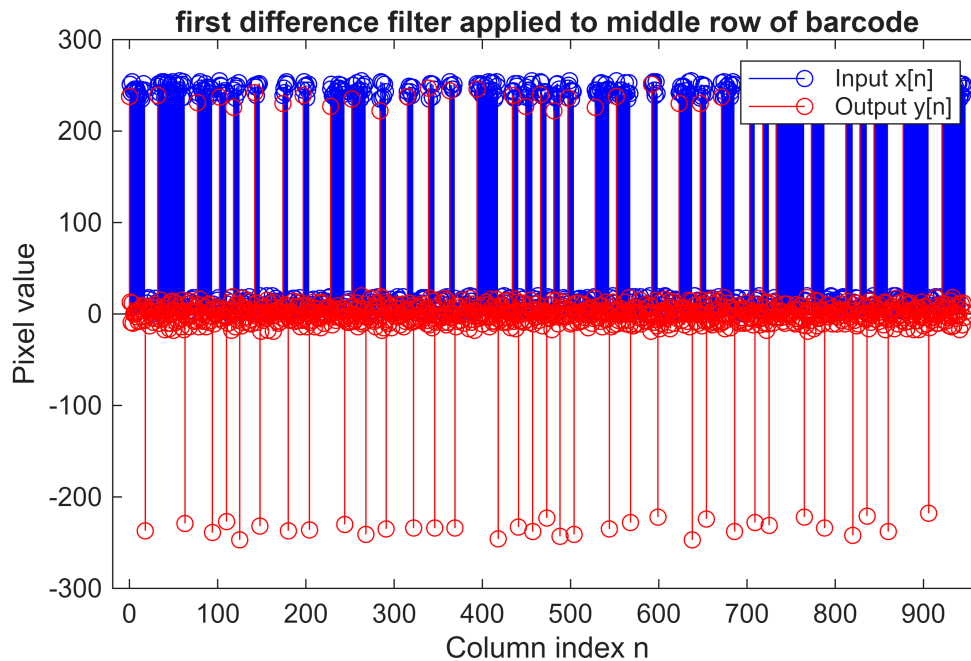
```
clc;
clear close all;
I = imread("HP110v3.png");
[rows, cols] = size(I);
mid = round(rows/2);
x = double(I(mid, :));
h = [1 -1];
y = conv(x,h);
y = y(1:length(x));
n = 0:length(x)-1;

figure
stem(n, x, 'b')
hold on
stem(n, y, 'r')
hold off
```

```

title('first difference filter applied to middle row of barcode')
xlabel('Column index n')
ylabel('Pixel value')
legend('Input x[n]', 'Output y[n]')

```



The spike magnitudes differ because the pixel values in the image are not perfectly 0 and 255. Due to noise s , the white pixels range from about 235 to 255 and the black pixels from about 0 to 20. Each spike in $y[n]$ is the difference between two neighboring pixels at an edge, so its size depends on exactly which values meet there. A bright white (255) next to a dark black (0) gives a spike near 255, while a duller white (235) next to a grayer black (20) gives only about 215.

```

tau = 200;
d = abs(y) >= tau;

```

The `find` function returns the indices (positions) of all non-zero elements in an array, not the values themselves. For a logical or 0/1 vector like d , it returns the positions where d is 1 (true). from which we will create our position signal l when will point to the every transition point between white and black region of barcode

```

l = find(d);
delta = conv(l, h);
delta = delta(2:length(l));
section = delta(6:64);

theta = sum(section)/95;

bar_widths = round(section/theta);

decodeUPC(bar_widths)

```

```
ssbeg = 1×3
      1      1      1
ans = 1×12
      8      8      2      7      8      0      4      5      0      1      6      5
```

**Assessment Rubric
Lab 7: Convolution-II**

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Marks distribution:

	Tasks	LR2 Code	LR5 Results	LR9 Report	AR4 Report Submission	AR7 Comments
In-Lab	Task 1		/8	/04	/02	
	Task 2	/8	/4			/2
	Task 3	/8	/4			/2
	Task 4	/12	/12			/4
Total =70		/28	/28	/04	/02	/8
CLO Mapped		CLO2	CLO2	CLO2	CLO3	CLO3

CLO – (LDL)	Total Points	Points Obtained
CLO2 – (Psy-4)	56	
CLO3 – (Aff-3)	14	
Total	70	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.



Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3: Good Points 3	Level 4: Exemplary Points 4
LR2	Program/Code/ Simulation Model/ Network Model	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR3	Troubleshooting	Unable to identify the fault/minimal effort show in troubleshooting.	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and takes necessary steps and actions to correct it.
LR4	Data Collection	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in using symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and task completed in due time.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR6	Calculations	Formulae used and/or calculations are incorrect. Units are not mentioned with results.	Formulae used are correct but there are few mistakes in calculation which lead to incorrect results.	Formulae used and end results are correct. Complete working steps are not shown. Proper units are not mentioned with results.	Formulae used, end results and all calculations are correct with all the intermediate steps clearly shown. Units are mentioned with results.
LR9	Report	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.
LR11	Design	Proposed design is unsubstantiated or does not satisfy most of given constraints. The breakdown of system into features is incomplete and still at lower resolution.	Justification for proposed design is weak, and it only satisfies some constraints. The breakdown of system is missing some features and resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.
AR4	*Report Submission	Not accepted after 1 week.	Late submission after 2 days and within 1 week.	Late submission after the lab timing and within 2 days of the due date.	Timely submission of the report and in the lab time.
AR7	Report Content/Code comments	Most of the questions are not answered / figures are not labelled/ titles are not mentioned / units are not mentioned. No comments are present in the code.	Some of the questions are answered, figures are labelled, titles are mentioned and units are mentioned. Few comments are stated in the code.	Majority of the questions are answered, figures are labelled, titles are mentioned and units are mentioned. Comments are stated in the code.	All the questions are answered, figures are labelled, titles are mentioned and units are properly mentioned. Proper comments are stated in the code.